

# Multiphase Oxidation of Sulfur Dioxide in Aerosol Particles: Implications for Sulfate Formation in Polluted Environments

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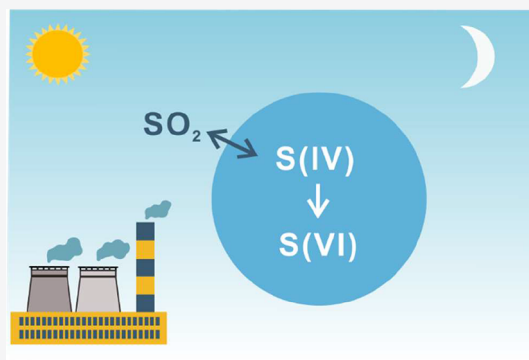


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**ABSTRACT:** Atmospheric oxidation of sulfur dioxide ( $\text{SO}_2$ ) forms sulfate-containing aerosol particles that impact air quality, climate, and human and ecosystem health. It is well-known that in-cloud oxidation of  $\text{SO}_2$  frequently dominates over gas-phase oxidation on regional and global scales. Multiphase oxidation involving aerosol particles, fog, and cloud droplets has been generally thought to scale with liquid water content (LWC) so multiphase oxidation would be negligible for aerosol particles due to their low aerosol LWC. However, recent field evidence, particularly from East Asia, shows that fast sulfate formation prevails in cloud-free environments that are characterized by high aerosol loadings. By assuming that the kinetics of cloud water chemistry prevails for aerosol particles, most atmospheric models do not capture this phenomenon. Therefore, the field of aerosol  $\text{SO}_2$  multiphase chemistry has blossomed in the past decade, with many oxidation processes proposed to bridge the difference between modeled and observed sulfate mass loadings. This review summarizes recent advances in the fundamental understanding of the aerosol multiphase oxidation of  $\text{SO}_2$ , with a focus on environmental conditions that affect the oxidation rate, experimental challenges, mechanisms and kinetics results for individual reaction pathways, and future research directions. Compared to dilute cloud water conditions, this paper highlights the differences that arise at the molecular level with the extremely high solute strengths present in aerosol particles.



## 1. INTRODUCTION

Multiphase oxidation of sulfur dioxide ( $\text{SO}_2$ ) in aerosol particles, fog, or cloud droplets plays an important role in global atmospheric chemistry, being involved in acid precipitation, haze pollution, and climate change. Indeed, one of the major accomplishments in atmospheric multiphase chemistry was the early demonstration that in-cloud processing of  $\text{SO}_2$  dominates over much slower gas-phase loss in many environments. It was shown that uptake of  $\text{SO}_2$  to aqueous hydrometeors that contain dissolved oxidants is important in driving acidic sulfate formation. The work on this topic from the past few decades is nicely documented in textbooks<sup>1,2</sup> and reviews in the literature.<sup>3–6</sup>

Although it is known that  $\text{SO}_2$  can be oxidized on the surfaces of mineral dust particles,<sup>7</sup> soot,<sup>8</sup> and ice,<sup>9</sup> the multiphase oxidation of  $\text{SO}_2$  in aqueous submicrometer aerosol particles has largely been considered to be a negligible sink process because the particulate liquid water content (LWC) is much lower than for fog or cloud liquid water.<sup>10</sup> As a result, chemical transport models systematically underestimate the conversion of  $\text{SO}_2$  to sulfate, on both global and regional scales.<sup>6,11</sup> The model-measurement disagreements for  $\text{SO}_2$  and sulfate are lessened when hypothetical heterogeneous processing of  $\text{SO}_2$  proceeds on aerosol particles at a rate comparable to the gas-phase oxidation of  $\text{SO}_2$  in the northern hemisphere summer.<sup>12</sup> It is also possible that multiphase processing of  $\text{SO}_2$  in aerosol particles

during long-range transport contributes to the sulfate burden in Arctic region.<sup>13</sup>

In recent years, scientific interest in the multiphase oxidation of  $\text{SO}_2$  in aqueous aerosol particles has been rekindled by the observation of rapid sulfate aerosol production during cloud-free, severe haze events in East Asia.<sup>14–18</sup> It is likely that such chemistry also occurs in other highly polluted environments where coal burning is widely prevalent, such as South Asia, although detailed field studies are still lacking in that regard. Figure 1a illustrates that a proxy of the sulfate formation rate, the ratio of sulfate to  $\text{SO}_2$ , is significantly higher during highly polluted periods ( $\text{PM}_{2.5} > 300 \mu\text{g m}^{-3}$ ) than during clean periods ( $\text{PM}_{2.5} < 50 \mu\text{g m}^{-3}$ ) in Beijing in 2013. Traditional atmospheric models that include gas-phase oxidation of  $\text{SO}_2$  by the hydroxyl radical ( $\text{OH}$ )<sup>19</sup> and stabilized Criegee intermediates<sup>16</sup> and a suite of aqueous, in-cloud  $\text{SO}_2$  oxidation processes cannot capture the high levels of aerosol sulfate during these haze events.<sup>15,20,21</sup> Good correlation between the unexplained

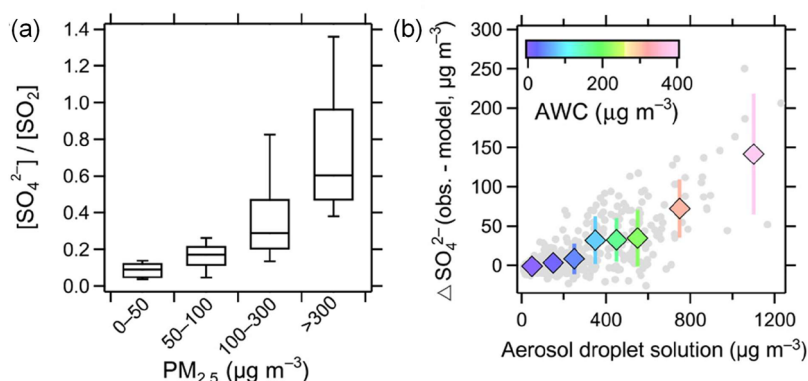
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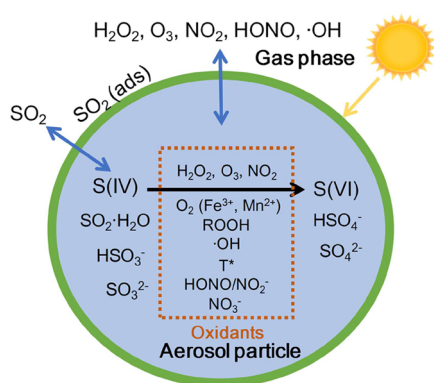




**Figure 1.** Characteristic features of a haze event in Beijing, China in January 2013. (a) Ratio of sulfate to  $\text{SO}_2$  as a function of particle mass loading ( $\text{PM}_{2.5}$ ). (b) Correlation of the difference between observed and modeled sulfate with the mass of aerosol droplet solution (aqueous aerosol) and aerosol water content (AWC). Taken from ref 16 under the terms of Creative Commons CC BY license.

sulfate content and aerosol LWC (Figure 1b), and the greatly improved model performance after including a hypothetical particle-mediated heterogeneous oxidation process, highlight the potentially important role of multiphase oxidation of  $\text{SO}_2$  in aerosol particles.<sup>21</sup>

Multiphase oxidation of  $\text{SO}_2$  in aerosol particles refers to the interaction between  $\text{SO}_2$  and aerosol particles, which can take place both at the surface and in the bulk aerosol water medium.<sup>22</sup> The conventional view of multiphase reaction processes includes the transfer of  $\text{SO}_2$  into the condensed aerosol phase, dissolution of hydrated  $\text{SO}_2$  into  $\text{H}^+$ ,  $\text{HSO}_3^-$ , and  $\text{SO}_3^{2-}$ , and oxidation of S(IV) ( $= \text{SO}_2 \cdot \text{H}_2\text{O} + \text{HSO}_3^- + \text{SO}_3^{2-}$ ) by oxidants that either dissolve from the gas phase or are generated in the aerosol phase to form S(VI) (Figure 2). Typical oxidants



**Figure 2.** Schematic of processes involved in multiphase oxidation of  $\text{SO}_2$  in aerosol particles. Green coating represents the potential for an organic-rich separated phase when liquid–liquid phase separation occurs, which could limit the mass transfer. The double arrow indicates the gas-particle partitioning of the reactants.

considered in models include hydrogen peroxide ( $\text{H}_2\text{O}_2$ ),<sup>23</sup>  $\text{O}_3$ ,<sup>23</sup>  $\text{O}_2$  catalyzed by transition metal ions (TMI, i.e., Fe (III) and Mn (II)),<sup>24–27</sup> methyl hydrogen peroxide,<sup>28</sup> and peroxyacetic acid.<sup>28</sup> To bridge the gap between modeled and measured sulfate levels, additional oxidants have recently been suggested to be nitrogen dioxide ( $\text{NO}_2$ ),<sup>14,16</sup> organic peroxides ( $\text{ROOH}$ ),<sup>29–32</sup> photolysis products of nitrate ( $\text{pNO}_3^-$ ),<sup>33–35</sup> and excited triplet states of photosensitizer molecules ( $\text{T}^*$ ).<sup>36</sup> Hypohalous acid ( $\text{HOX}$ , e.g.,  $\text{HOCl}$  and  $\text{HOBr}$ ), two important S(IV) oxidants in the marine atmosphere, may also be important under specific haze conditions considering their large rate

constants with S(IV) and high water solubility.<sup>37</sup> In addition to the reactions that occur with molecules fully dissolved in the particles,  $\text{SO}_2$  may also adsorb and react on the aqueous aerosol particle surface.<sup>38</sup>

A complete understanding of the multiphase oxidation of  $\text{SO}_2$  in aerosol particles requires detailed knowledge of the above processes. However, the complexity and varying properties of aerosol particles hinder accurate determination of related kinetics and their incorporation into the atmospheric models. In particular, the major factor that distinguishes chemistry occurring in cloud droplets and aerosol particles is the exceedingly high particulate solute concentrations. Not only can this lead to strong radical sources within the particle (e.g., from nitrate photolysis or photosensitized reactions; see below) but also to fast radical sinks as well. Also, these high solute concentrations can make the particles viscous and affect mass transfer rates of  $\text{SO}_2$  and oxidants, and they can lead to high ionic strengths that can have strong effects on the dissolution of S(IV) into its ionic forms ( $\text{HSO}_3^-$  and  $\text{SO}_3^{2-}$ ) and on the rate constants of the elementary reactions that involve ionic mechanisms.<sup>39</sup> Detailed treatment of these solute strength effects in multiphase chemistry models is lacking for not only  $\text{SO}_2$  oxidation processes but many other multiphase reactions.<sup>40</sup> For simplicity, enhancement factors of the reaction rate in aerosol particles relative to the dilute solution can be used to describe the overall effects of the solute strength.<sup>41</sup>

In this review article, we summarize recent advances in multiphase oxidation of  $\text{SO}_2$  in aerosol particles, with a focus on laboratory studies performed since 2013 when severe haze pollution in China was first documented in detail. Prior work has been reviewed in previous publications,<sup>3,5,42</sup> but this does not capture the explosion of activity that has happened in the past few years. We first discuss the factors affecting the  $\text{SO}_2$  oxidation rate and significant experimental challenges that arise in characterizing the reaction kinetics. We then present a summary of the mechanisms and kinetics results for a series of individual reaction pathways of current interest. We conclude with recommendations for future research. A theme running throughout the article is the importance of contrasting results measured on aerosol particles to those from bulk solution studies which represent cloud water conditions.  $\text{SO}_2$  oxidation processes provide a valuable testbed for assessment of how accurately results from cloud water oxidation experiments can be extrapolated to the very different chemical environments of aqueous aerosol particles. This review article will not cover the

current state and knowledge of modeling as they have been well treated by Ervens.<sup>6,43</sup>

## 2. FACTORS AFFECTING SO<sub>2</sub> OXIDATION RATE

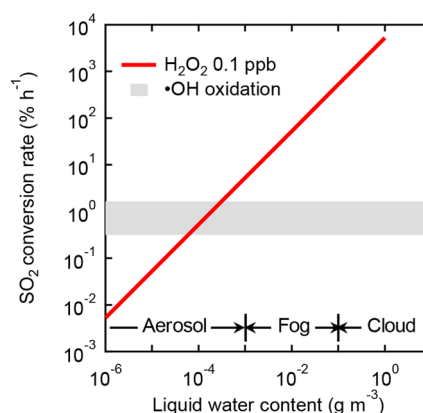
The physical (e.g., size and LWC) and chemical properties (e.g., pH and solute strength) of aqueous aerosol particles are significantly different from those of fog or cloud droplets. A detailed overview of the properties of aerosol particles, fog, and cloud droplets in different environments can be found in a previous review.<sup>39</sup> Supporting Information (SI) Table S1 summarizes the LWC, pH, and ionic strength of atmospheric cloud, fog, and aerosol particles in different environments. The aerosol LWC varies from tens of  $\mu\text{g m}^{-3}$  in the typical urban atmosphere to hundreds of  $\mu\text{g m}^{-3}$  under severe haze conditions, which are several orders of magnitude lower than  $\sim 0.1 \text{ g m}^{-3}$  for fog/cloud droplets. Aqueous aerosol particles behave as nonideal concentrated solutions with ionic strengths of  $>10 \text{ mol kg}^{-1}$ , which is significantly higher than  $\sim 10^{-4} \text{ mol kg}^{-1}$  for fog/cloud droplets. For aerosol particles consisting of a single compound, the ionic strength typically shows a negative correlation with the aerosol LWC (SI Figure S1). However, their relationships would be more complicated in the atmosphere because their values are highly dependent on the RH and chemical compositions of ambient aerosols. Currently, aerosol pH is estimated based on thermodynamic models because direct measurement of pH is challenging.<sup>44</sup> According to current understanding, aerosol particles in urban areas are expected to be acidic with pH ranging from  $-1$  to  $5$ . Fog and cloud droplets are also generally acidic, but to a significantly lesser degree than aerosol particles.<sup>45</sup>

**2.1. Liquid Water Content.** The LWC plays an important role in determining the aqueous oxidation rate of gas phase SO<sub>2</sub> ( $R_a$ ) which can be seen from the rate expression written for the case of an oxidant also present in the gas phase:<sup>1</sup>

$$R_a (\% \text{ h}^{-1}) = 3.6 \times 10^{-10} k H_{\text{oxi}} H_{\text{SO}_2}^* \xi_{\text{oxi}} L R T \quad (1)$$

where  $k$  is the reaction constant in  $\text{M}^{-1} \text{ s}^{-1}$ ,  $H_{\text{oxi}}$  and  $H_{\text{SO}_2}^*$  are Henry's law constants of the oxidant and SO<sub>2</sub> in  $\text{M atm}^{-1}$ , respectively.  $\xi_{\text{oxi}}$  is the mixing ratio of the oxidant in ppb,  $L$  is the liquid water content in  $\text{g m}^{-3}$ ,  $R = 0.082 \text{ atm L K}^{-1} \text{ mol}^{-1}$ , and  $T$  is in K.

To illustrate, Figure 3 shows the calculated 298 K conversion rate of SO<sub>2</sub> oxidation by 0.1 ppb H<sub>2</sub>O<sub>2</sub> in aerosol particles, fog, and cloud at pH 3 as a function of LWC, using the reaction mechanism, rate constant and partitioning parameters appropriate for dilute solutions as present in cloud water (see details in SI Tables S2 and S3). In Section 4 below, it is demonstrated that these parameters are not necessarily appropriate for the high solute strengths in particulate matter. As well, the figure shows the rate of oxidation by gas phase OH radicals present at  $1\text{--}5 \times 10^6 \text{ molecules cm}^{-3}$ . As expected, the aqueous SO<sub>2</sub> conversion rate increases linearly with the LWC. Compared to oxidation by the gas phase OH radical, aqueous oxidation in fog and cloud droplets dominates the SO<sub>2</sub> conversion rate. On a global scale, in-cloud multiphase oxidation of SO<sub>2</sub> was modeled to contribute approximately 60% of sulfate production, which is three times that of gas-phase oxidation.<sup>26</sup> Though the SO<sub>2</sub> conversion rate at low aerosol LWC is negligible, it can be important at high aerosol LWC conditions. An interesting feature of this chemistry is that oxidation to sulfate within an aerosol particle raises the amount of hygroscopic material in that particle which increases



**Figure 3.** Conversion rate of SO<sub>2</sub> oxidation by 0.1 ppb gas-phase H<sub>2</sub>O<sub>2</sub> in aerosol particles, fog, and cloud at pH 3 as a function of liquid water content<sup>2</sup> and by gas phase OH radical of  $1\text{--}5 \times 10^6 \text{ molecules cm}^{-3}$  at 298 K. The H<sub>2</sub>O<sub>2</sub> solubility is determined by Henry's Law constant to water.

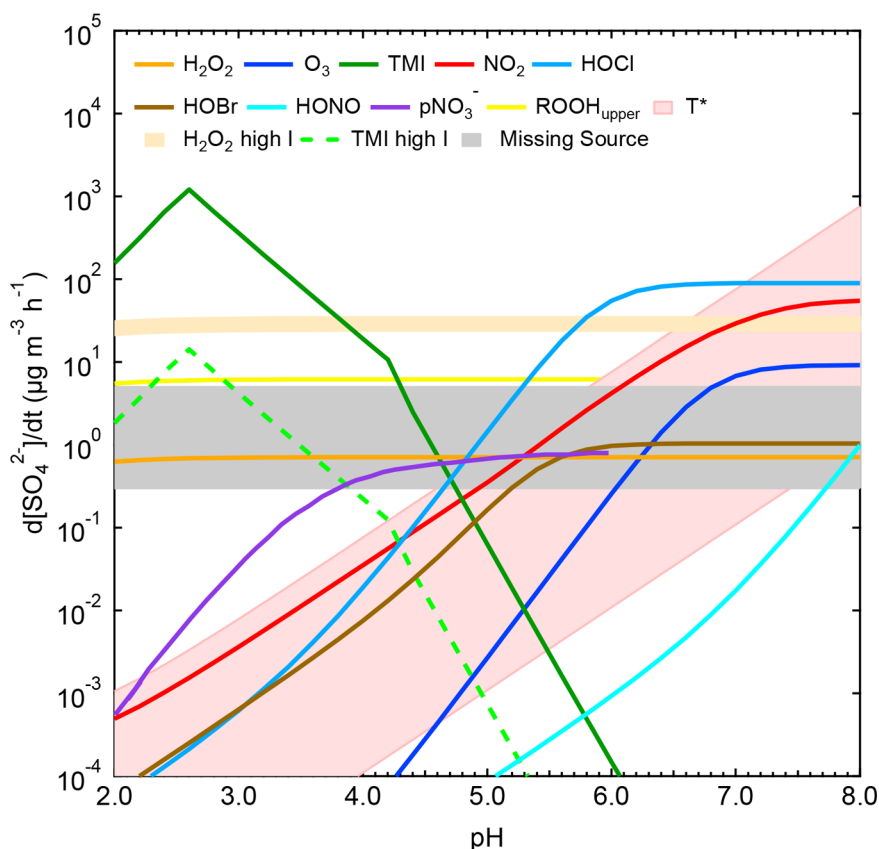
its LWC as well, i.e. there is some degree of autocatalysis that proceeds.<sup>46</sup>

**2.2. pH Effects.** The solubility and effective Henry's law constant of SO<sub>2</sub> are positively dependent on pH, resulting in more dissolved S(IV) (mostly HSO<sub>3</sub><sup>−</sup> at pH of 2–7) in water,<sup>1</sup> which will enhance the sulfate formation rates for many aqueous SO<sub>2</sub> oxidation pathways involving O<sub>3</sub>, NO<sub>2</sub>, nitrous acid (HONO), HOCl, HOBr, and T\*. At the same time, hydrogen ions (H<sup>+</sup>) catalyze the aqueous SO<sub>2</sub> oxidation pathways involving H<sub>2</sub>O<sub>2</sub> and ROOH, which will offset the effects arising from pH-dependent solubility of SO<sub>2</sub> and lead to pH-independent SO<sub>2</sub> oxidation rates for the H<sub>2</sub>O<sub>2</sub> and ROOH pathways for pH conditions above 2. Consequently, aerosol pH is a critical factor that affects the sulfate formation rate in aerosol water. In particular, at moderate aerosol pH (4–5) of China winter haze (see ref 47 and references therein), many of the aqueous SO<sub>2</sub> oxidation pathways potentially have atmospheric importance.

To illustrate the pH effects for different multiphase SO<sub>2</sub> oxidation pathways in aerosol particles, we calculate the instantaneous sulfate formation rates under urban winter haze conditions, following the method of Cheng et al.<sup>16</sup> (Figure 4). The rate expressions and parameters used are appropriate for bulk, low concentration solutions (SI Table S3), with no considerations for high solute strength effects. The characteristic conditions are representing Beijing winter haze conditions and taken from refs 16, 29, 35, 36, and 48–51. It is seen that S(IV) oxidation by H<sub>2</sub>O<sub>2</sub>, ROOH, O<sub>2</sub> catalyzed by TMI, NO<sub>2</sub>, HOCl, pNO<sub>3</sub><sup>−</sup>, and T\* in aerosol particles may be important while HOBr and HONO may not be, if bulk phase kinetics are applicable to aerosol particles. A number of new findings regarding these oxidation pathways question this assumption (see Section 4).

**2.3. Solute Strength Effects.** As emphasized earlier, aerosol particles are characterized by very high ionic strength (e.g., arising from the presence of ammonium, sulfate, nitrate, etc.) and by high concentrations of organic molecules (e.g., organic acids). The particle may be phase separated with water-insoluble organics partitioning into a more hydrophobic component and water-soluble organics combining with inorganic ions in a water-rich hydrophilic fraction.<sup>52</sup> For reaction to proceed, hydration of SO<sub>2</sub> occurs first, followed by acidic dissociation of sulfurous acid into HSO<sub>3</sub><sup>−</sup> and SO<sub>3</sub><sup>2−</sup>.





**Figure 4.** Sulfate formation rates by multiphase  $\text{SO}_2$  oxidation in aerosol particles under urban winter haze conditions. The characteristic data conditions for generating the colored solid lines were taken from refs 16, 29, 35, 36, and 48–51 [ $\text{SO}_2(\text{g})$ ] = 40 ppb, [ $\text{H}_2\text{O}_2(\text{g})$ ] = 0.1 ppb,<sup>48</sup> [ $\text{NO}_2(\text{g})$ ] = 66 ppb, [ $\text{O}_3(\text{g})$ ] = 1 ppb, [ $\text{HOCl}(\text{g})$ ] = 1 ppt,<sup>49</sup> [ $\text{HOBr}(\text{g})$ ] = 0.1 ppt,<sup>50</sup> [ $\text{HONO}(\text{g})$ ] = 9 ppb,<sup>51</sup> pH-dependent concentrations of [Fe(III)] and [Mn(II)],<sup>16</sup> photolysis rate of nitrate ( $\text{pNO}_3^-$ ) =  $10^{-5} \text{ M s}^{-1}$ ,<sup>35</sup> [ $\text{ROOH}(\text{aq})$ ] =  $5 \times 10^{-3} \text{ M}$ ,<sup>29</sup> [ $\text{T}^*(\text{aq})$ ] =  $2.3 \times 10^{-13}$ – $1.6 \times 10^{-10} \text{ M}$ ,<sup>36</sup> AWC =  $300 \mu\text{g m}^{-3}$ , aerosol droplet radius  $R_p = 0.15 \mu\text{m}$ ,  $T = 271 \text{ K}$ . The accommodation coefficient values for  $\text{SO}_2$ ,  $\text{H}_2\text{O}_2$ ,  $\text{O}_3$ ,  $\text{NO}_2$ ,  $\text{HOCl}$ ,  $\text{HOBr}$ , and  $\text{HONO}$  are 0.11, 0.23,  $2 \times 10^{-3}$ ,  $2 \times 10^{-4}$ ,<sup>16</sup> 0.8,<sup>153</sup> 0.8,<sup>154</sup> and 0.05,<sup>2</sup> respectively. The light orange-shaded area and dashed green line represent the sulfate formation rates for  $\text{H}_2\text{O}_2$  and TMI reaction pathways after accounting for the impact of ionic strength, respectively.<sup>61</sup> The gray-shaded area shows typical East Asian missing sulfate formation rates of  $0.3$ – $5 \mu\text{g m}^{-3} \cdot \text{h}^{-1}$  as a reference.<sup>16</sup>

Thus, it is likely that the reaction proceeds largely in the water-rich fraction. However, the chemical environment of this fraction is significantly different from that in pure dilute solutions that approximate cloud water. As a result, the  $\text{SO}_2$  oxidation rate in aerosol particles has the potential to be either enhanced or reduced relative to the rate in dilute solutions.

First, the electrolytes may enhance (“salting in”) or reduce (“salting out”) the partitioning of  $\text{SO}_2$  and gas-phase oxidants to the water-rich phase, depending on the composition and concentration of the solutes. Salting effects are usually described by the Sechenov equation:<sup>53</sup>

$$\log(H^0/H) = K_s c_s \quad (2)$$

which relates the ratio of Henry’s law constant of gas dissolved in the absence,  $H^0$ , and presence,  $H$ , of a given concentration of salt,  $c_s$ .  $K_s$  is the Sechenov coefficient, which is specific to both the gas and salt. Schumpe and co-workers<sup>54,55</sup> have developed a method to obtain  $K_s$  parameters. However, these parameters have been largely developed from measurements in bulk solutions which cannot approach the high solute concentrations present in aerosol particles that are thermodynamically metastable at low relative humidity (RH).<sup>56,57</sup> Although solute concentration-dependent solubilities of  $\text{SO}_2$  and  $\text{H}_2\text{O}_2$  were determined for sodium chloride (NaCl) solutions mimicking marine aerosols,<sup>58,59</sup> there are no data available regarding the

salting effects of continental aerosol particles on the solubility of  $\text{SO}_2$  and its oxidants.

Second, the ionic strength impacts the dissociation of sulfurous acid. For example, Millero et al.<sup>59</sup> observed that the stoichiometric dissociation constants of sulfurous acid first increase and then decrease with ionic strength ( $<6 \text{ mol kg}^{-1}$ ) in NaCl solutions. Reliable dissociation constants of sulfurous acid are needed to accurately estimate the concentrations of S(IV) species and thus the pH-dependent sulfate formation rate. To estimate the stoichiometric dissociation constant in aerosol particles, it is necessary to estimate the activity coefficient of the solutes. Though several methods have been developed to estimate the activity coefficients of the solutes in mixed electrolyte solutions, the interactions of inorganic and organic electrolytes are not well understood, hindering reliable estimates of activity coefficients of S(IV) species at high ionic strengths.<sup>60</sup>

Third, high ionic strength affects the rate constants for the reactions that drive aqueous S(IV) oxidation.<sup>39</sup> Previous experimental investigations on ionic strength effects on reaction rate constants of some aqueous S(IV) oxidation pathways are limited for ionic strengths below  $5 \text{ mol kg}^{-1}$ .<sup>16,41</sup> Only a few recent studies have investigated the high ionic strength effects (see Section 4).<sup>61</sup>

Finally, the organics present in the solution may participate in the S(IV) oxidation reactions even if they are not the principal

oxidants. For example, general acid catalysis effects can accelerate the reaction between S(IV) and H<sub>2</sub>O<sub>2</sub> by acting as proton donors<sup>61</sup> but inhibit the TMI-catalyzed S(IV) oxidation by complexation of the TMI.<sup>62</sup> However, the effects have not been studied for the vast majority of other organics in the particles and for different reaction systems.

**2.4. Mass Transfer Limitations.** Via both eddy and molecular diffusion, SO<sub>2</sub> and its gas-phase oxidants need to transfer to the surface of the particle to initiate oxidation. Gas-phase mass transfer can be rate limiting for uptake of gases to cloud droplets, which can be 10 μm in size, when the SO<sub>2</sub> (or oxidant) uptake coefficient is above 0.1.<sup>63</sup> However, for submicron aerosol particles with diameters closer to the mean free path of the molecules, this process is too fast to slow down the reaction.

A second limitation can occur associated with a small mass accommodation coefficient, which is defined as the fraction of collisions with a surface that leads to uptake to that surface. The mass transfer rate coefficient ( $k_{MT}$ ) can be described using a resistor model:<sup>1</sup>

$$k_{MT} = \left[ \frac{R_p^2}{3D_g} + \frac{4R_p}{3\alpha\nu} \right]^{-1} \quad (3)$$

where  $R_p$  is the aerosol radius,  $D_g$ ,  $\alpha$ , and  $\nu$  are the gas-phase molecular diffusion coefficient, mass accommodation coefficient, and mean molecular speed of a given gas. As shown in Figure 4, mass accommodation limitations occur for some aqueous SO<sub>2</sub> oxidation pathways such as NO<sub>2</sub>, O<sub>3</sub>, HOCl, and HOBr at pH above 6.5 for 150 nm particles. Given that the aerosol pH of East Asian winter haze is typically 4–5 (see ref 47 and refs therein), this mass transfer limitation can be considered to be negligible for SO<sub>2</sub> oxidation.

Lastly, the rate of mass transfer can be slow within an aerosol particle if the viscosity is high. This can occur with organic aerosol particles, especially at low temperatures and relative humidities. However, if an aerosol particle phase separates into organic-rich and water-rich components and forms partially engulfed morphologies, the aqueous fraction will likely have low viscosity with fast diffusion rates for solvated reactants. The small size of submicrometer particles also makes it unlikely for diffusion within the particle to be rate limiting. By contrast, a core–shell morphology that has a viscous organic shell coating on the aqueous core may limit the rate of mass transfer.

An extreme case of low viscosity within particles arises for solids. In addition, SO<sub>2</sub> may adsorb and then react on the surfaces of solid particles such as mineral dust and soot particles to form sulfate. A quantitative understanding of the rates for these surface reactions is complex because the rates highly depend on the physical and chemical nature of the surfaces, including surface area, surface defects, and surface absorbed water.<sup>2</sup>

**2.5. Interfacial Chemistry.** An additional important distinction between cloud and aerosol chemistry is that the surface area to volume ratios of aerosol particles is higher than that of cloud droplets, so that chemistry that proceeds faster at the gas-particle interface will be of greater relative importance. While it is known that S(IV) oxidation proceeds on some solid substrates<sup>7,8</sup> and that undissociated SO<sub>2</sub> adsorbs to water surfaces,<sup>38,64,65</sup> it is as yet inconclusive whether the rates of chemistry at the aqueous interface are of atmospheric importance. For example, one report for the reaction H<sub>2</sub>O<sub>2</sub> and SO<sub>2</sub> under cloud water conditions concludes that a bulk

description of the kinetics fully describes experimental observations in a droplet train flow reactor.<sup>66</sup> However, there are new results described below for the O<sub>2</sub><sup>67,68</sup> and formaldehyde (HCHO)<sup>69</sup> oxidant systems that are not commensurate with bulk phase chemistry and may be occurring at the interface. For example, the presence of a SO<sub>2</sub>–water complex at the interface opens up the potential for interface-mediated photochemistry.<sup>70</sup>

### 3. EXPERIMENTAL REQUIREMENTS FOR STUDYING SO<sub>2</sub> OXIDATION KINETICS IN AEROSOL PARTICLES

Given that solute concentrations can be much higher in aerosol particles than in bulk solutions, the optimal experimental approach for laboratory SO<sub>2</sub> oxidation kinetics studies employs particles in an aerosol flow tube, environmental chamber, or optical trap. Thermodynamically metastable solutions with high concentrations can be sustained in aerosol particles because liquid-to-solid nucleation rates are slower than in bulk solutions.<sup>71</sup> An added advantage of working with aerosol particles is that mass transfer constraints are significantly reduced, with rapid diffusion in both the gas and condensed phases.<sup>72</sup>

Aerosol flow tubes allow control of variable reaction time, ease of wall conditioning to gas phase species, and convenient assessment of background conditions, that is, measurement of changes in gas phase or aerosol composition when one reaction partner is not present. The major limitation to aerosol flow tubes is short reaction times, so that results must be extrapolated to ambient concentration conditions from the higher concentrations used in the experiment. By contrast, in environmental chambers reactant concentrations can be much closer to those in the environment because reaction times are longer. In particular, environmental chamber experiments are suitable to study the kinetics of SO<sub>2</sub> oxidation by NO<sub>2</sub> because the relatively low concentrations of NO<sub>2</sub> used can reduce the aerosol acidity changes due to the hydrolytic disproportionation of NO<sub>2</sub> on the aerosol surface.<sup>73</sup> However, assessing wall losses is much more challenging. Longer time-scale experiments are also possible using optical traps coupled to suitable analytical techniques to characterize sulfate formation, such as Raman spectroscopy. Mass transfer limitations may occur with the optical trap method as the particle size is typically 1–10 μm.

Aerosol kinetics experiments are best configured with simultaneous measurements of gas-phase reactant concentrations and the rate of sulfate formation in the aerosol particles. If the experiment is operated by measuring SO<sub>2</sub> decay then the background decay rate in the absence of particles must be assessed, that is, loss of SO<sub>2</sub> that occurs on flow tube or chamber walls. Care should be taken to do these measurements at the appropriate conditions given that this decay can be RH-dependent. As well, the decay rate with particles present should scale linearly with total particle volume or surface area. Experiments can be configured with SO<sub>2</sub> in excess and the formation rate of sulfate is measured instead; this reduces complications that arise if there is nonparticle-driven loss of SO<sub>2</sub>. The use of aerosol mass spectrometry is particularly convenient for the ability to monitor particle sulfate levels in a quantitative manner so long as the sulfate detection efficiency is characterized.<sup>74</sup>

Sulfate formation rates are sensitive to changes in aerosol pH arising from the formation of sulfuric acid but few experiments control aerosol pH because it is experimentally challenging. In one study, pH control was achieved using pH-buffered,

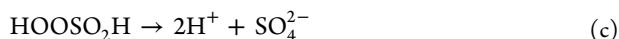
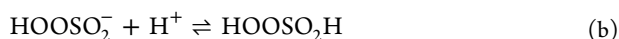
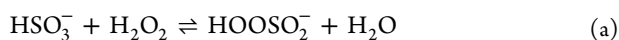
submicrometer, deliquesced aerosol particles.<sup>61</sup> In combination with thermodynamically based models,<sup>56,57</sup> this approach allows the isolation of the pH and ionic strength effects of aerosol particles.

#### 4. LABORATORY STUDIES OF REACTION PATHWAYS OF CURRENT INTEREST

This section gives an overview of the mechanistic and analytic information on a series of individual SO<sub>2</sub> oxidation pathways in aerosol particles, involving peroxides, O<sub>2</sub>, reactive nitrogen, photochemical reactions, hypohalous acids, aldehydes, and mineral dust. All these pathways may be important in polluted environments.

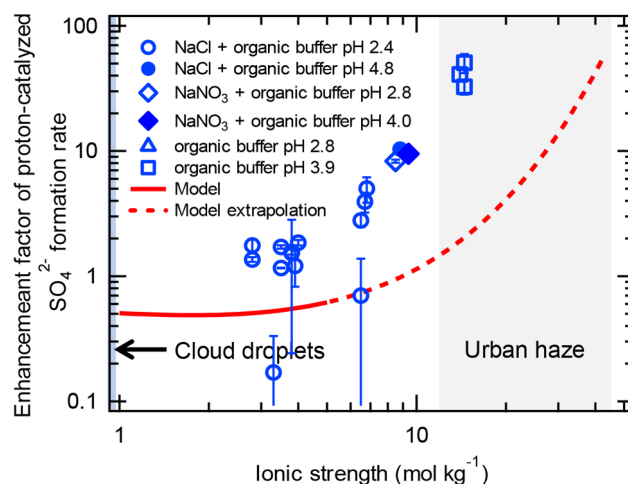
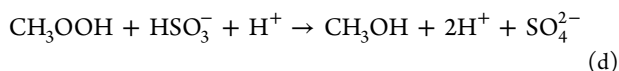
**4.1. Peroxides.** Peroxides are reactive compounds that arise from HO<sub>2</sub> and RO<sub>2</sub> association reactions in the gas phase, auto-oxidation processes,<sup>75</sup> Criegee intermediate reactions, and to some degree from ozonolysis of alkenes. H<sub>2</sub>O<sub>2</sub> is a highly water-soluble compound whereas organic peroxides are a major component of secondary organic aerosol.<sup>76</sup>

The kinetics and mechanism of S(IV) oxidation by H<sub>2</sub>O<sub>2</sub> in dilute solutions have been well investigated.<sup>23</sup> It is generally accepted that this reaction proceeds through a nucleophilic reaction between HSO<sub>3</sub><sup>−</sup> and H<sub>2</sub>O<sub>2</sub> to form a peroxymonosulfurous acid intermediate, which undergoes acid-catalyzed rearrangement to form sulfuric acid:<sup>77</sup>



Recent advances have been made regarding the particle ionic strength effects on this reaction system. Ali et al.<sup>58</sup> have studied the reaction kinetics in deliquesced sodium chloride particles using a cross-flow minireactor for ionic strengths up to 5 mol kg<sup>−1</sup>. The reaction rates can be well predicted using the empirical relationships between ionic strength and Henry's law constants of SO<sub>2</sub> and H<sub>2</sub>O<sub>2</sub>, the dissociation constant of sulfurous acid, and the literature rate constant. However, the extrapolation of these empirical relationships to higher ionic strength conditions may not be appropriate. Liu et al.<sup>61</sup> extended the ionic strength effects to ~14 mol kg<sup>−1</sup> by investigating the kinetics in deliquesced malonic acid buffer aerosol particles in an aerosol flow tube. They found an enhancement factor of 33–51 in the kinetics at an ionic strength of 14 mol kg<sup>−1</sup> compared to the rate in the dilute solutions (Figure 5). This enhancement may arise from the ionic strength effect on the dissociation of the peroxymonosulfurous acid intermediate and potentially the salting-in of reactants. As well, the malonic acid buffer can act as a general acid and accelerate the kinetics of the reaction. As shown in Figure 4, the sulfate formation rate for the H<sub>2</sub>O<sub>2</sub> pathway under urban winter haze conditions is greatly enhanced after accounting for the ionic strength effects. Moreover, the aerosol phase H<sub>2</sub>O<sub>2</sub> concentration can be significantly higher than that predicted from partitioning of gas phase H<sub>2</sub>O<sub>2</sub>,<sup>78</sup> which would further increase the sulfate formation rate.

In addition to H<sub>2</sub>O<sub>2</sub>, organic hydroperoxides (ROOH) have also been found to oxidize SO<sub>2</sub> in the aqueous phase.<sup>1</sup> For example, methylhydroperoxide (CH<sub>3</sub>OOH) reacts with dissolved HSO<sub>3</sub><sup>−</sup> through



**Figure 5.** Enhancement factor of proton-catalyzed sulfate formation rate for multiphase oxidation of SO<sub>2</sub> by H<sub>2</sub>O<sub>2</sub> in aerosol particles as a function of ionic strength. Experiments were conducted for a mixture of NaCl and organic buffer aerosol particles (circle symbols), NaNO<sub>3</sub> and organic buffer aerosol particles (rhombus symbols) and organic buffer aerosol particles at aerosol pH of 2.8 (triangle symbols) and 3.9 (square symbols). The model results (solid line) describe the effects of ionic strength on the proton-catalyzed sulfate formation rate measured in bulk solutions.<sup>61</sup> The model extrapolation is shown as a dashed red line. Light blue- and gray-shaded areas indicate characteristic ionic strength ranges for cloud droplets and aerosol particles during urban haze episodes, respectively. All error bars represent one standard deviation. Reprinted in part with permission from ref 61. Copyright 2020 Proceedings of the National Academy of Sciences.

Because the kinetics and mechanism of aqueous oxidation of SO<sub>2</sub> by small ROOH species (e.g., CH<sub>3</sub>OOH and peroxyacetic acid) are relatively well understood,<sup>1</sup> recent research has focused more on the oxidation by multifunctional ROOH, for example, organic peroxides derived from monoterpenes<sup>29–31</sup> and isoprene.<sup>32</sup> For example, Ye et al.<sup>31</sup> demonstrated the reactivity of ROOH from ozonolysis of  $\alpha$ -pinene and limonene toward SO<sub>2</sub> at high RH through SO<sub>2</sub> uptake experiments in an environmental chamber. However, it is difficult to determine the rate constant between ROOH and S(IV) in aerosol particles due to the challenge in accurately quantifying the aerosol ROOH content. Yao et al.<sup>29</sup> estimated that the second-order reaction rate constant of S(IV) with  $\alpha$ -pinene-derived peroxides in aerosol particles is several times lower than the bulk values of H<sub>2</sub>O<sub>2</sub><sup>23</sup> and isomers of isoprene hydroxyl hydroperoxide (ISOPOOH),<sup>32</sup> but significantly higher than those of 2-butanone and *tert*-butyl hydroperoxide.<sup>30</sup> In particular, Wang et al.<sup>30</sup> implemented the rate constants between S(IV) and ROOH standards into a box model, Gas Aerosol Model for Mechanism Analysis (GAMMA), to predict the SO<sub>2</sub> decay for the chamber experiment<sup>31</sup> involving limonene SOA and SO<sub>2</sub>. They found that the reaction between SO<sub>2</sub> and ROOH dominates the SO<sub>2</sub> decay during the experiment. Dovrou et al.<sup>32</sup> modeled that sulfate formation via cloud processing of SO<sub>2</sub> oxidation by ISOPOOH may be important in regions with high isoprene emissions and low NO<sub>x</sub> conditions.

To illustrate the importance of multiphase oxidation of SO<sub>2</sub> by ROOH in aerosol particles, we calculate the upper limit of the sulfate formation rate for the ROOH pathway under urban winter haze conditions by adopting the rate constant from Yao et al.<sup>29</sup> and assuming that the reaction rate expression of ROOH



and S(IV) is similar to that of  $\text{H}_2\text{O}_2$  and S(IV) (Figure 4). This pathway is of atmospheric importance because of the high ROOH content in aerosol particles, driven by these species' high solubility and polarity. However, large uncertainties in the above estimation arise from the lack of pH- and ionic strength-dependent reaction rate constants between ROOH and S(IV) in aerosol particles as well as the ROOH content. Moreover, this reaction forms organosulfates via mechanisms that are not fully understood and the yield of inorganic sulfate varies among different ROOH.<sup>29–32</sup>

In summary, currently available evidence suggests that the high ionic strength of aerosol particles enhances the sulfate formation rate via  $\text{H}_2\text{O}_2$  oxidation. Future research is needed to explore the empirical expression of the ionic strength-dependent reaction rate in aerosol particles. The kinetics and mechanisms of S(IV) oxidation by ROOH in aerosol particles are not clear for peroxides formed from biogenic precursors and even less well understood for peroxides arising from oxidation of anthropogenic volatile organic compounds (VOCs). No studies have investigated the ionic strength effects on the reactions between S(IV) and ROOH. Given that many large ROOH compounds, at least those with only a few functional groups,<sup>79</sup> may reside in the organic-rich component of phase-separated aerosol particles, the relevance of reactions conducted in the organic phase needs to be assessed.

**4.2.  $\text{O}_2$ .** Previous studies have shown that in-cloud S(IV) oxidation by dissolved  $\text{O}_2$  catalyzed by TMI plays an important role in global sulfate formation.<sup>26,27</sup> As well, this reaction in aerosol particles is proposed to be important during urban winter haze events using the parameters obtained for dilute solutions.<sup>16</sup> However, this reaction is a complex and poorly constrained process because the kinetics and mechanism are highly dependent on the environmental conditions of pH, S(IV) concentrations and ionic strength.<sup>4,5</sup> For example, the reaction rate for Mn(II)-catalyzed S(IV) oxidation is more than 100 times slower at an ionic strength of  $1 \text{ mol kg}^{-1}$  than the rate in a dilute solution that approximates cloud water conditions.<sup>1</sup> Moreover, it is found that organic molecules can significantly inhibit the reaction rates, probably by complexation of the transition metal ion.<sup>1,62</sup> Lastly, this reaction can occur on the surfaces of solid particle that may have significant levels of sorbed water.<sup>80–82</sup> We note that the dissolved  $\text{O}_2$  is typically in excess and is not likely to limit the rate of this reaction.

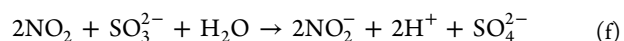
Studies regarding the ionic strength effects on TMI-catalyzed  $\text{SO}_2$  oxidation in aerosol particles remain scarce. Liu et al.<sup>61</sup> investigated the ionic strength effects using a mixture of deliquesced malonic acid buffer, NaCl, iron(III) chloride ( $\text{FeCl}_3$ ), and manganese(II) chloride ( $\text{MnCl}_2$ ) aerosol particles in an aerosol flow tube. It was found that the sulfate formation rate is suppressed by a factor of approximately 85 at an ionic strength of  $2.8 \text{ mol kg}^{-1}$  compared to dilute solutions. This effect can be well described by the extended Debye–Hückel equation. As a result, the sulfate formation rate for the TMI pathway under urban winter haze conditions may be significantly reduced after accounting for the ionic strength effects (Figure 4).

Uncatalyzed oxidation of  $\text{SO}_2$  by  $\text{O}_2$  can also occur but the rate has generally believed to be negligible.<sup>1</sup> However, Hung and co-workers recently investigated  $\text{SO}_2$  oxidation by  $\text{O}_2$  on the surface of microdroplets using a spray-chamber reactor coupled to an electrospray ionization mass spectrometer.<sup>67,68</sup> They concluded that the rates via surface reactions at pH values below 3.5 are 1–2 orders magnitude higher than that of  $\text{SO}_2$  oxidation by  $1 \text{ ppb } \text{H}_2\text{O}_2$  under typical atmospheric conditions. However,

implementation of this pathway in a global chemical transport model (GEOS-Chem) leads to an unrealistically high contribution (95%) of this pathway to the global tropospheric sulfate formation.<sup>50</sup> Additionally, if this process proceeded with the rate determined by Hung and co-workers,<sup>67,68</sup> more than 100 times faster sulfate formation rate should have been observed for the TMI experiments performed at aerosol pH of 2.8 by Liu et al.<sup>61</sup> Clearly, more work is needed to elucidate the role of uncatalyzed  $\text{SO}_2$  oxidation on acidic aerosol surfaces.

Soot particles may play an important role in catalytic oxidation of  $\text{SO}_2$  by  $\text{O}_2$  to form sulfate.<sup>83</sup> Recently, using in situ attenuated total internal reflection infrared spectroscopy, He et al.<sup>8</sup> demonstrated that soot particles can catalyze the oxidation of  $\text{SO}_2$  by  $\text{O}_2$  to form sulfate. They also found that high RH accelerates the reaction rate. A density functional theory calculation from the same group<sup>84</sup> proposed that water can dissociate into surface hydroxyl groups and promote the conversion of  $\text{SO}_2$  to  $\text{SO}_3$  on soot surfaces. Correspondingly, the surface composition of soot can impact  $\text{SO}_2$  oxidation. For example, Zhao et al.<sup>85</sup> found that the reactivity of  $\text{SO}_2$  toward soot surfaces depends on the amounts of aromatic C–H and oxygen-containing groups on the soot surfaces.

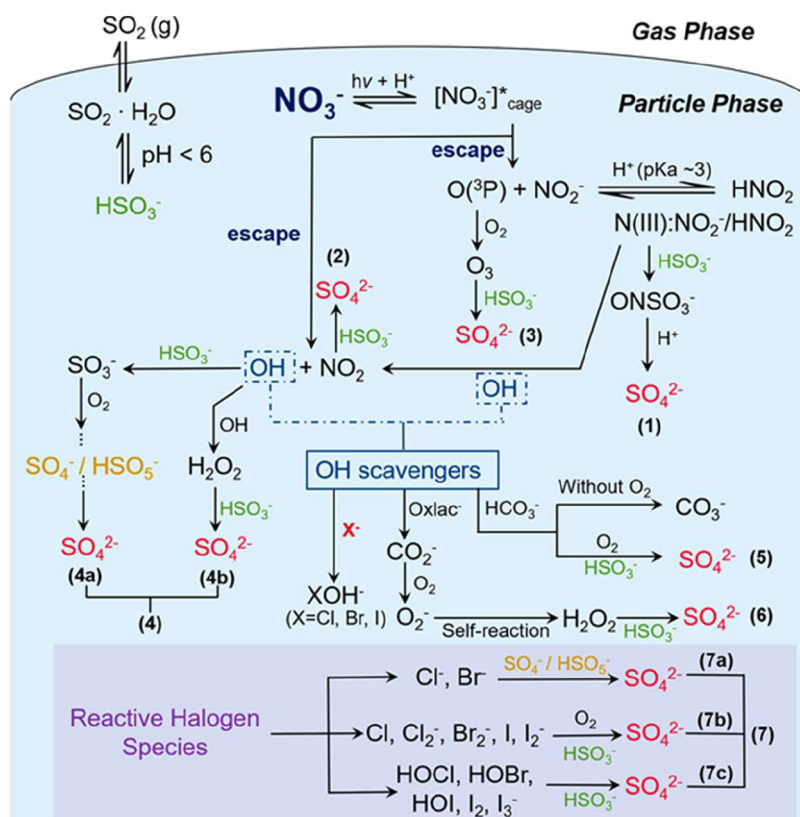
**4.3. Reactive Nitrogen.**  $\text{NO}_2$  and HONO are two major reactive nitrogen species that can oxidize  $\text{SO}_2$  in the aqueous phase.  $\text{SO}_2$  oxidation by both  $\text{NO}_2$  and HONO in fog droplets<sup>51,86–88</sup> and aerosol particles<sup>14,16</sup> has been proposed to contribute to the missing sulfate under East Asian urban winter conditions. The overall reactions can be expressed as



However, large uncertainties may arise from uncertainties in kinetic data obtained in dilution solutions. For example, Clifton et al.<sup>89</sup> reported 1–2 orders of magnitude higher rate constants for the  $\text{NO}_2$  reaction with  $\text{SO}_2$  than that of Lee and Schwartz<sup>90</sup> at similar pH values. The results of Clifton et al.<sup>89</sup> may be overestimated due to the interference of a complex product on the  $\text{NO}_2$  decay measurement.<sup>91</sup>

The atmospheric importance of  $\text{SO}_2$  oxidation by  $\text{NO}_2$  in aerosol particles under urban winter haze conditions is unclear. Wang et al.<sup>14</sup> reported reactive uptake coefficients ( $\gamma_{\text{SO}_2}$ ) of  $0.8\text{--}3.9 \times 10^{-4}$  for  $\text{SO}_2$  when exposing oxalic acid seed aerosols to high mixing ratios of  $\text{SO}_2$  (250 ppb),  $\text{NO}_2$  (250 ppb), and ammonia ( $\text{NH}_3$ ) (1 ppm) at high RH in an environmental chamber. They concluded that this process is feasible with high RH and  $\text{NH}_3$  neutralization and is key to the efficient sulfate formation under urban winter haze conditions. But according to past mechanistic work, this pathway should become important only when the aerosol pH value is above 5.5,<sup>14,16</sup> which is higher than the values inferred from thermodynamic models (see ref 47 and refs therein).

By contrast, another study from the same group observed negligible sulfate formation when exposing ammonium sulfate seed aerosols to vapors of  $\text{SO}_2$ ,  $\text{NO}_2$ , and  $\text{NH}_3$  at high RH.<sup>92</sup> The distinct results between the studies with oxalic acid and ammonium sulfate seed aerosols are attributed to the potentially different aerosol pH values of the two experimental systems. In particular, the pH may be 1 unit higher with  $0.1 \text{ mol kg}^{-1}$  ammonium oxalate bulk solutions compared to the same concentration of ammonium sulfate solutions. One concern is that these pH differences in bulk solutions may not directly translate to aerosol particles because the excess gas phase



**Figure 6.** Mechanisms for multiphase oxidation of  $\text{SO}_2$  by nitrate photolysis at 300 nm in the presence of halide ions or OH scavengers ( $\text{NaHCO}_3$  and oxalic acid). Reprinted from ref 33. Copyright 2020 American Chemical Society.

ammonia in chamber experiments plays an important role in setting aerosol pH and may drive the aerosol pH of the two systems to similar levels.

Another concern is that the ionic strengths of the two reaction systems are different. In another system, Chen et al.<sup>93</sup> investigated the ionic strength effects on  $\text{SO}_2$  oxidation by  $\text{NO}_2$  in the presence of  $\text{NH}_3$  in oxalic acid aerosol particles at 98% RH in a smog chamber. It was found that the rate constant is enhanced by a factor of more than 100 at an ionic strength of  $1 \text{ mol kg}^{-1}$  compared to the dilute solution with ionic strengths of  $2.4\text{--}7.2 \times 10^{-6} \text{ mol kg}^{-1}$ . However, this enhancement effect of ionic strength is inconsistent with the slow sulfate formation observed in the ammonium sulfate particle (ionic strength  $>9 \text{ mol kg}^{-1}$ ) experiments of Wang et al.<sup>92</sup> This indicates that ionic strength may not play an important role in the multiphase oxidation of  $\text{SO}_2$  by  $\text{NO}_2$ . Some other factors, such as interfacial reactions,<sup>94</sup> may be more important.

The mechanism of  $\text{SO}_2$  oxidation by  $\text{NO}_2$  in aerosol particles is not fully understood. It is assumed that gas phase  $\text{NO}_2$  dissolves in aerosol water as it does in bulk solutions. However, Li et al.<sup>73</sup> reported that more than 95% of  $\text{NO}_2$  disproportionates to form HONO and  $\text{HNO}_3$  at the surface of aqueous sodium bisulfite ( $\text{NaHSO}_3$ ) microjets rather than oxidizing dissolved  $\text{HSO}_3^-$ . In contrast, Ge et al.<sup>95</sup> found that the  $\text{NO}_2$  reaction with  $\text{SO}_2$  in deliquesced  $\text{NaCl}$  aerosol particles in the presence of  $\text{NH}_3$  dominates over the  $\text{NO}_2$  disproportionation.

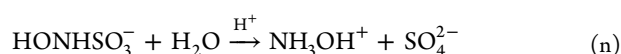
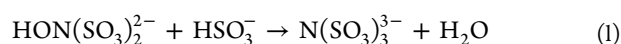
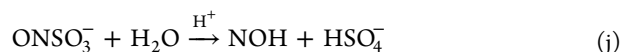
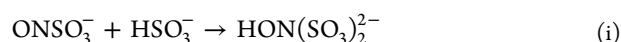
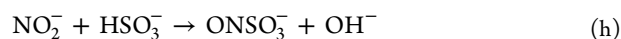
Multiphase reactivity of  $\text{SO}_2$  and  $\text{NO}_2$  on the surface of mineral dust is another topic of current interest (SI Table S4). This process is influenced by  $\text{O}_2$ ,<sup>96,97</sup>  $\text{NH}_3$ ,<sup>98</sup> and RH.<sup>99,100</sup> For example, Zhao et al.<sup>99</sup> reported a  $\gamma_{\text{SO}_2}$  of  $\sim 10^{-8}$  for  $\text{SO}_2$  reaction with  $\text{NO}_2$  on  $\text{CaCO}_3$  particles using  $\text{N}_2$  as the carrier gas in a flow

cell coupled with Micro-Raman spectroscopy, while the  $\gamma_{\text{SO}_2}$  increases to  $\sim 10^{-5}$  using  $\text{O}_2$  as the carrier gas.<sup>97</sup> It is suggested that  $\text{NO}_2$  initiates radical formation and  $\text{O}_2$  acts as the main oxidant, consistent with the findings of He et al.<sup>96</sup> The proposed initiation mechanism is the following:

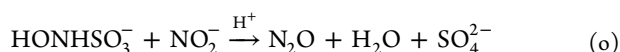


$\text{NH}_3$  was found to promote sulfate formation on acidic  $\alpha\text{-Fe}_2\text{O}_3$  and  $\alpha\text{-Al}_2\text{O}_3$  surfaces while this effect is weaker on basic  $\text{CaO}$  and  $\text{MgO}$  surfaces possibly due to the basic nature of  $\text{CaO}$  and  $\text{MgO}$ .<sup>98</sup> The extrapolation of laboratory parameters to atmospheric conditions is very challenging because the physical and chemical nature of the surfaces used in laboratory studies can be quite different from those in ambient conditions.

Although not well understood, the mechanisms of aqueous oxidation of dissolved  $\text{SO}_2$  by  $\text{HONO}/\text{NO}_2^-(\text{N(III)})$  have been proposed to be<sup>101</sup>





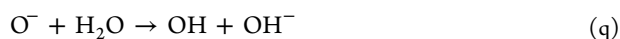


Studies on multiphase oxidation of  $\text{SO}_2$  by HONO in aerosol particles are scarce. Li et al.<sup>73</sup> proposed that N(III) species may drive the multiphase oxidation of  $\text{SO}_2$  in aerosol particles during urban winter haze. However, adopting past studies of the kinetics of  $\text{SO}_2$  oxidation by  $\text{NO}_2^-$  in bulk solutions, Ge et al.<sup>95</sup> estimated that N(III) oxidation leads to minor sulfate production compared to the  $\text{NO}_2$  oxidation pathway for environmental chamber experiments involving  $\text{SO}_2$ ,  $\text{NO}_2$ ,  $\text{NH}_3$ , and deliquesced NaCl particles.

By contrast, Zhang et al.<sup>102</sup> investigated  $\text{SO}_2$  reaction with  $\text{NO}_2$  on black carbon (BC) particles in the presence of  $\text{NH}_3$  in an environmental chamber. They found that HONO produced from the  $\text{NO}_2$  reaction with BC facilitates sulfate formation. A parametrization based on the chamber results predicts more than 30% of sulfate formation from this BC-catalyzed pathway during urban haze events.

**4.4. Photochemical Reactions.** In highly polluted environments with the solar dimming effect from high aerosol particle loading, photochemical effects can be reduced in importance compared to more pristine environments.<sup>103</sup> However, even in polluted winter haze conditions in East Asia, recent observations have demonstrated that photochemistry is more important than previously thought.<sup>104,105</sup> As a result, a number of recent studies have focused on multiphase photochemical oxidation of  $\text{SO}_2$ , involving  $\text{pNO}_3^-$ ,<sup>33–35</sup>  $\text{T}^*$ ,<sup>36</sup> and photochemical reactions on the surface of mineral dust.<sup>106,107</sup>

One focus has been upon the kinetics and mechanism of multiphase oxidation of  $\text{SO}_2$  by photolysis of nitrate in aerosol particles, based on past studies that have demonstrated that reactive species form from this photochemical process with actinic radiation:



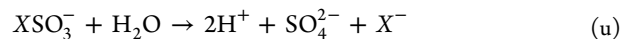
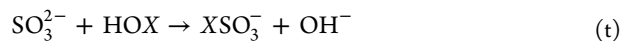
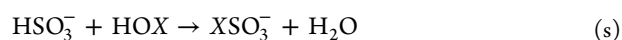
Gen et al.<sup>34,35</sup> first demonstrated sulfate production from  $\text{pNO}_3^-$  at irradiation of 250 nm and then extended their studies to atmospherically relevant wavelengths of 300 nm in a flow cell probed by Raman spectroscopy. For photolysis experiments conducted at 300 nm, the sulfate production was predicted to be dominated by N(III) formed from the photolysis of nitrate.<sup>35</sup> The  $\gamma_{\text{SO}_2}$  was found to be insensitive to aerosol pH and slightly negatively dependent on RH while positively correlated with the photon flux. As well, this reaction is not suppressed by the presence of organics while it is accelerated by halide ions which are thought to lead to less hydration of nitrate.<sup>33</sup> The overall mechanisms are shown in Figure 6. Figure 4 illustrates that sulfate formation from  $\text{pNO}_3^-$  can be important under urban winter haze conditions assuming a high rate of nitrate photolysis ( $= j_{\text{NO}_3^-} \times [\text{NO}_3^-]$ ) of  $10^{-5} \text{ M s}^{-1}$ .<sup>35</sup> Zheng et al.<sup>108</sup> implemented this pathway into the Community Multiscale Air Quality (CMAQ) model and estimated that it contributes between 15% and 65% of the difference between modeled and observed sulfate concentrations during urban winter haze. The large uncertainty arises from the uncertainties in the reaction kinetics. In addition to nitrate photolysis, it was found that photolysis of iron-chloride containing aerosol particles produces

$\text{Cl}\bullet/\text{Cl}_2\bullet^-$  that also lead to rapid multiphase oxidation of  $\text{SO}_2$ , with a  $\gamma_{\text{SO}_2}$  of  $\sim 10^{-5}$ .<sup>109</sup>

Photosensitization chemistry is an emerging field in aerosol chemistry<sup>110</sup> that is founded in concepts explored originally in the natural waters photochemistry community. Solar radiation excites a photosensitizing molecule to a triplet state that can react with another molecule through energy transfer, electron transfer, or hydrogen atom abstraction mechanisms. Recent aqueous solution experiments<sup>36,111</sup> and aerosol flow tube experiments<sup>36</sup> have demonstrated that photosensitizer proxies of humic-like substances (HULIS) can promote  $\text{SO}_2$  oxidation to sulfate. For example, it was found that the reaction rate constants for S(IV) and  $\text{T}^*$  are large, ranging from  $6.9 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$  to  $1.0 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$  for 4-(benzoyl)benzoic and xanthone, typical photosensitizers.<sup>36</sup> In particular, Wang et al.<sup>36</sup> determined the reaction rate constant for S(IV) and  $\text{T}^*$  species extracted from ambient filters collected during haze events through pulsed laser excitation experiments. Applying this kinetic parameter to urban winter haze conditions results in a fast but uncertain sulfate formation rate at aerosol pH of 4–5 by assuming that the rate constant is pH-independent (Figure 4). It has not been determined whether this reaction proceeds through energy transfer, electron transfer, or hydrogen atom abstraction mechanisms. Moreover, the effects of pH and ionic strength on this reaction remain undetermined.

Several studies have attempted to investigate the kinetics of multiphase photochemical oxidation of  $\text{SO}_2$  on surfaces of mineral dust (SI Table S4). For example, Park and Jang<sup>107</sup> found that the  $\gamma_{\text{SO}_2}$  for Arizona Test Dust (ATD) particles in the presence of UV light is 1 order of magnitude higher than under dark conditions in chamber experiments. As well, the  $\gamma_{\text{SO}_2}$  value is enhanced by RH and the presence of other oxidants such as  $\text{O}_3$  and  $\text{NO}_2$ .<sup>106,107</sup> In contrast, the uptake coefficient of  $\text{SO}_2$  on  $\text{TiO}_2$  surfaces under UV irradiation conditions decreases with the increase of RH.<sup>112</sup> The proposed mechanism for multiphase photochemical oxidation of  $\text{SO}_2$  on surfaces of mineral dust involves electron–hole pair generation from metal oxides followed by oxidation and reduction processes (ref 87 and references therein).

**4.5. Hypohalous Acid.** It is well-known that HOX (where X = Cl, Br, and I) can oxidize S(IV) in aqueous solutions.<sup>113</sup> It is generally accepted that the reaction occurs via nucleophilic attack of the electron pair on the sulfur atom of  $\text{HSO}_3^-$  and  $\text{SO}_3^{2-}$  at the halide atom of HOX to form the  $\text{XSO}_3^-$  intermediate, followed by its rapid hydrolysis to form  $\text{SO}_4^{2-}$ :<sup>114</sup>



Mechanistically, the rate constants of HOX with  $\text{SO}_3^{2-}$  are higher than those of HOX with  $\text{HSO}_3^-$ , consistent with the higher nucleophilicity of  $\text{SO}_3^{2-}$  than  $\text{HSO}_3^-$ .<sup>115</sup> As well, the reactivity of HOX follows the order of  $\text{HOI} > \text{HOBr} > \text{HOCl}$ , reflecting the ability of HOX to more easily transfer  $\text{X}^+$  to the nucleophiles.<sup>116</sup>

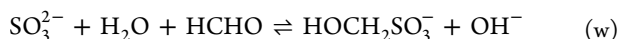
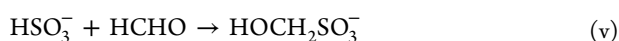
Although multiphase oxidation of  $\text{SO}_2$  by HOX in aerosol particles has been studied to better understand the pristine marine environment,<sup>117</sup> it may be important under haze conditions in coastal urban environments. As well, there is growing evidence for substantial levels of active halogen species present at considerable distances removed from marine

sources.<sup>118119120121</sup> For example, halogens are known to be emitted from coal burning,<sup>122</sup> salt lakes,<sup>123</sup> and road salt.<sup>124</sup>

Recently, Liu and Abbatt<sup>37</sup> measured the rate constants of HOCl and HOBr with  $\text{HSO}_3^-$  using deliquesced malonic acid buffer aerosol particles in an aerosol flow tube. Prior to these measurements, model studies have assumed that the rate constants of HOCl and HOBr with  $\text{HSO}_3^-$  are the same as those for reactions with  $\text{SO}_3^{2-}$ .<sup>113114125</sup> Chen et al.<sup>114</sup> modeled that the in-cloud S(IV) oxidation by HOBr contributes 8% of global sulfate production, and Hoffmann et al.<sup>125</sup> found that including multiphase halogen chemistry in a model results in a 9% increase of in-cloud sulfate formation in polluted coastal areas. Applying the new kinetic parameters to urban winter haze conditions illustrates a potentially important role of HOCl in forming sulfate assuming 1 ppt of HOCl is present (Figure 4). Given that the HOCl mixing ratio can reach 300 ppt in eastern China, future model studies should consider multiphase oxidation of  $\text{SO}_2$  by HOCl in aerosol particles.<sup>121</sup> HOBr is less important than HOCl in forming sulfate assuming 0.1 ppt HOBr during urban winter haze events. Large uncertainty remains in this estimation due to the lack of observational constraints on the HOBr mixing ratios. Additionally, HOI may also play a role in forming sulfate in polluted coastal cities. However, the reaction rate constant of S(IV) oxidation by HOI is unknown.

Overall, a comprehensive understanding of the importance of HOX in sulfate formation requires a better knowledge of the kinetics of aqueous S(IV) oxidation by HOX. In particular, the effects of organics on multiphase oxidation of  $\text{SO}_2$  by HOX in aerosol particles should be investigated.

**4.6. Aldehydes.** In addition to the oxidation chemistry that proceeds with organic peroxide species, other organic molecules that are part of polluted aerosol could be reactive toward  $\text{SO}_2$ . Whereas reactions with carboxylic acid, alcohol, or ester functional groups are unlikely, dissolved S(IV) reacts with aldehydes, most notably formaldehyde, to form hydroxymethanesulfonate (HMS):



In the case of polluted environments, it has been suggested that traditional ion chromatography methods misinterpret HMS as sulfate.<sup>126127128</sup> The production of HMS is favored under moderate pH because the decomposition rate of HMS increases more rapidly than its formation rate with the increase of pH. Song et al.<sup>126</sup> found that HMS may account for up to one-third of the unexplained sulfate during urban winter haze events. At the same time, the calculated HMS production rate in aerosol particles at pH 5.5 based on bulk kinetics of HCHO and  $\text{SO}_2$  was consistent with the measurement. It was also found that this reaction can proceed at the gas–water interface, but with dissolved HCHO concentrations greatly in large excess of those that will prevail in the atmosphere.<sup>69</sup>

Other aldehydes can also react with  $\text{SO}_2$  to produce other hydroxyalkylsulfonate species that may be atmospherically important. For example, De Haan et al.<sup>129</sup> observed the formation of HMS,  $\text{C}_2$ , and  $\text{C}_3$  sulfonates from the reaction between S(IV) and glyoxal in bulk aqueous solutions and aerosol particles at atmospherically relevant pH values.

**4.7. Mineral Dust.** As mentioned earlier,  $\text{SO}_2$  can be oxidized on the surfaces of mineral dust particles. These reactions have been reviewed by Usher et al.<sup>7</sup> The reaction mechanism is complex, strongly dependent on the environ-

mental conditions (e.g., RH) and the physical and chemical nature of mineral dust particles. Given that mineral dusts can have many monolayers of water sorbed to their surfaces,<sup>7</sup> similar chemistry may proceed in these solute-rich surface films as occurs with aerosol particles. In addition, high RH conditions can promote  $\text{SO}_2$  oxidation because surface absorbed water can facilitate  $\text{SO}_2$  absorption.<sup>80130131</sup>

Recent studies have investigated  $\text{SO}_2$  uptake on mineral dust proxy particles<sup>80132133134135136137</sup> and authentic mineral dust particles<sup>130131138</sup> as well as the effects of nitrate,<sup>131133</sup>  $\text{H}_2\text{O}_2$ ,<sup>130135</sup>  $\text{NH}_3$ ,<sup>136</sup>  $\text{NO}$ ,<sup>137</sup> acetaldehyde,<sup>134</sup> RH,<sup>80130</sup> and mineral dust structure (SI Table S4).<sup>80</sup> For example, nitrate was found to enhance the uptake of  $\text{SO}_2$  on the surface of hematite via in situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) and a White cell coupled with Fourier transform infrared spectroscopy (white cell-FTIR)<sup>133</sup> and ATD particles using flow tube experiments.<sup>131</sup> Kong et al.<sup>133</sup> suggested that the enhancement is due to the formation of HONO that further oxidizes  $\text{SO}_2$  to sulfate while Zhang et al.<sup>131</sup> attributed the enhancement to the formation of  $\text{HNO}_3$  that accelerates the radical-chain reaction of  $\text{SO}_2$  oxidation.  $\text{H}_2\text{O}_2$ ,<sup>130135</sup>  $\text{NH}_3$ ,<sup>136</sup> and  $\text{NO}$ <sup>137</sup> were also found to enhance the  $\text{SO}_2$  uptake on mineral dust particles.

## 5. SUMMARY, RECOMMENDATIONS, AND OUTLOOK

In studies of East Asian urban haze, it has been observed that sulfate formation occurs when large amounts of secondary organic aerosol, BC,  $\text{NO}_x$  and HONO,  $\text{NH}_3$ , VOCs, and  $\text{SO}_2$  are present.<sup>139</sup> The meteorological conditions are frequently cool, with temperatures sometimes around freezing, and humid (RH of ~40–90%).<sup>103139</sup> Light and oxidant (e.g.,  $\text{O}_3$ ) levels can be suppressed. These conditions are not unique to East Asia and may prevail in other polluted urban regions with a large amount of coal burning, such as South Asia.

Extensive recent laboratory studies described above have transformed our understanding of the aerosol multiphase chemistry of  $\text{SO}_2$ , with many new oxidation pathways that need to be considered. For example,  $\text{SO}_2$  oxidation by  $\text{H}_2\text{O}_2$  in aerosol particles may be important not only because the high ionic strength of aerosol particles enhances the sulfate formation rate<sup>61</sup> but also because the aerosol phase  $\text{H}_2\text{O}_2$  concentration can be significantly higher than that predicted from partitioning of gas phase  $\text{H}_2\text{O}_2$ .<sup>78</sup> Future studies into  $\text{H}_2\text{O}_2$  sources are warranted given that the rate of sulfate formation may be oxidant limited. As well, organic hydroperoxide molecules can be important oxidants of  $\text{SO}_2$  given that the peroxide content of aerosol particles is high in most polluted environments.<sup>140141</sup> While large uncertainties remain regarding transition metal ion solubility and aerosol concentrations, isotope-specific field measurements indicate an important role of TMI-catalyzed S(IV) oxidation.<sup>50142</sup> More work needs to be conducted on the mixing state of particles that contain mineral dust sources of these TMI, that is, whether the ions coexist with other reactants in individual particles. Motivated by high levels of  $\text{NO}_x$  in urban environments, laboratory experiments have shown that  $\text{SO}_2$  oxidation by  $\text{NO}_2$  in aerosol particles could become important at high aerosol pH values, but a major issue is whether such alkaline particles are widely present.<sup>1495</sup> It is not known whether  $\text{NO}_2$  molecules react predominantly through disproportionation to form nitrite and nitrate, or directly with S(IV).<sup>7395</sup> Lastly, photochemically driven oxidation of dissolved  $\text{SO}_2$  could be important during the daytime; however, more work is needed to better translate lab results to ambient conditions given the

widely different light absorption properties of laboratory photosensitizing agents and absorbing molecules in ambient aerosol.<sup>108</sup>

Sulfate oxidation rates in aerosol particles cannot yet be accurately predicted for all ambient conditions. Nevertheless, the new studies highlight mechanisms that are likely to be most important in different parts of the world. For example, in polluted environments of East Asia, the  $\text{H}_2\text{O}_2$  and organic peroxide pathways must be considered, especially if the  $\text{H}_2\text{O}_2$  formation rate is sufficient. Likewise, oxidation by high  $\text{NO}_x$  and TMI from mineral dust may prevail for conditions where the aerosol particles are, respectively, especially alkaline or acidic. If the haze is highly pronounced so that light levels are low, the photochemically driven pathways will be less important. The processes occurring in East Asia are likely to also proceed in polluted South Asian environments but the HOCl pathway may be important considering the high aerosol pH<sup>143</sup> and substantial levels of chlorine species detected in New Delhi.<sup>144</sup> By contrast, in the marine boundary layer, the TMI and organic peroxide pathways will be much less important given their low atmospheric loadings. The dominant oxidants may shift from HOCl and  $\text{O}_3$  for freshly emitted high pH sea-salt aerosols to  $\text{H}_2\text{O}_2$  for acidified sea-salt aerosol particles.

Several recommendations are made for future research:

**Establish Solute Strength-Dependent Kinetics.**  $\text{SO}_2$  multiphase chemistry models need to account for the effects that extremely high solute strengths have on the kinetics. For example, in cloudwater the ionic strength is typically  $\sim 10^{-4}$  mol  $\text{kg}^{-1}$  but values are between  $\sim 10$  and  $50$  mol  $\text{kg}^{-1}$  in urban haze particles. The importance of ionic strength on multiphase chemistry of this type has been made previously.<sup>145</sup> Recent studies have demonstrated that these effects can significantly enhance the rate of  $\text{SO}_2$  oxidation by  $\text{H}_2\text{O}_2$  while inhibiting the rate of TMI-catalyzed  $\text{SO}_2$  oxidation.<sup>61</sup> More laboratory studies are necessary to investigate the impact of solute strength on the reactions between  $\text{SO}_2$  and ROOH,  $\text{NO}_2$ , HONO, hypohalous acids, and aldehydes.

**Measure pH-Dependent Reaction Rate Constants.** Aerosol pH is very important. As illustrated by Figure 4,  $\text{NO}_2$  oxidation of S(IV) can be fast if aerosol particle pH is sufficiently high. Present understanding is that particle pH can be elevated when enhanced levels of ammonia (and potentially carbonates) are present,<sup>45,47</sup> but not sufficiently so to actually drive  $\text{SO}_2$  oxidation at an environmentally important rate.<sup>146</sup> And yet, laboratory studies observe fast sulfate formation when gaseous  $\text{NO}_2$  is exposed to seed aerosol, so long as ammonia/ammonium is present. This is motivation for tighter modeling and experimental constraints on aerosol acidity, with application to understanding the pH-dependent rate constants for S(IV) oxidation by  $\text{NO}_2$ , ROOH derived from biogenic and anthropogenic VOCs, and  $\text{T}^*$ .

**Understand the Role of Organics.** The varied roles of organics have to be fully evaluated. Highly soluble organic molecules may be internally mixed with hygroscopic inorganic ions, leading to enhanced liquid water content. And, organic molecules may act directly as reactants with S(IV), as in the case of organic peroxides and aldehydes. Organics acids can catalyze the reaction of  $\text{H}_2\text{O}_2$  with dissolved S(IV), as in the case of malonic acid.<sup>61</sup> They are also known to complex transition metal ions and affect the TMI reactivity.<sup>62</sup>

**Investigate the Impact of Particle Morphology.** It is important to understand the impact of liquid–liquid phase separation, where an inorganic-rich, aqueous phase separates

from a less polar, organic phase. Separated liquid phases may have distinct pH values,<sup>147</sup> and reactants such as organic peroxides may or may not be in close contact with dissolved S(IV) species. S(IV) oxidation may occur directly with organic constituents of the separated organic phase.

**Additional Constraints.** Given that most laboratory model systems do not yet sufficiently match ambient aerosol chemical properties and morphology, a general conclusion from this review is that it is premature to conclusively state that any one reaction mechanism or set of mechanisms can accurately describe the rate of sulfate formation that occurs via aerosol processing in polluted environments. To illustrate, consider the experiments involving  $\text{NO}_2$  as an oxidant<sup>147,3</sup> where there is not yet quantitative agreement for the rates of such chemistry at atmospherically relevant aerosol particle pH.

The community should work toward a consensus for the rates of these processes under realistic atmospheric conditions. In particular, there is need for additional constraints on the chemistry beyond only the sulfate formation rate. For example, kinetic studies of multiphase oxidation of  $\text{SO}_2$  in ambient aerosol particles will be insightful, as used to assess the kinetics of  $\text{N}_2\text{O}_5$  and  $\text{ClNO}_2$  uptake to tropospheric aerosol.<sup>148,149,150,151</sup> For example, the difference between field- and lab-derived uptake coefficients of  $\text{ClNO}_2$  and  $\text{N}_2\text{O}_5$  can indicate particulate chloride mass,<sup>148</sup> and particle phase state and organic content.<sup>151</sup> Lab-in-the-field kinetics studies of  $\text{SO}_2$  multiphase oxidation will help evaluate whether lab parametrizations are applicable to more complex ambient conditions.

Also, laboratory studies should be conducted under conditions matching those in the environment as closely as possible with tight constraints on ionic strength, phase state, and pH. Top-down constraints, such as isotope-specific field measurements coupled with modeling studies and new kinetic data, will provide useful information on understanding the relative importance of different S(IV) oxidation pathways. In addition, field measurements of tracers such as organosulfates can be used to constrain the contribution of the ROOH pathway to sulfate formation.

**Understand Global and Regional Scale Effects.** On regional and global scales, it will be useful to perform sensitivity analyses to assess whether the various aerosol oxidation pathways improve model-measurement agreement for seasonally dependent sulfate levels, wet deposition rates, and  $\text{SO}_2$ -to-sulfate ratios. For example, observations and a GEOS-Chem model study over the eastern U.S. indicate that wintertime sulfate formation is limited by the availability of oxidants.<sup>152</sup> This highlights the importance of oxidant concentrations, because oxidant increases may offset  $\text{SO}_2$  emission reductions. As well, studies in highly polluted environments other than East Asia are important to do. Given that anthropogenic sulfur levels will decline as coal become a less favored energy source, additional emphasis should be given to assessing sulfate formation from dimethyl sulfide oxidation mechanisms in the marine boundary layer, for example by the hypohalous acids in marine aerosol.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.0c06496>.

Calculation of the oxidation rate of gas phase  $\text{SO}_2$  (Text S1), properties of atmospheric cloud, fog, and aerosol



particles (Table S1), equilibrium constants (Table S2), aqueous reaction rate expressions and rate coefficients (Table S3), previous studies of SO<sub>2</sub> oxidation on the surface of mineral dust (Table S4), and correlations between ionic strength and aerosol LWC (Figure S1) (PDF)

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### Notes

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